

3

Emerging Trends

3.1 INTRODUCTION

The 21st century has been defined by application of and advancement in information technology. Information technology today has become an integral part of our daily life. It has served as a big change agent in different aspects of business and society. It has proven a game changer in resolving economic and social issues. Advancements in and application of information technology are ever changing.

A number of technologies are introduced almost every day. While some of these do not succeed and fade away over time, some prosper and persist, gaining attention of the users. There is hardly any field of human life that is not affected by information technology. Various technological trends like cloud computing, mobile computing, social media, ubiquitous computing, etc., are growing rapidly. Cloud computing enables us to share hardware and software resources as a service over the internet with any user across the globe on pay-per-use basis. Mobile computing helps us to access and process data on handheld mobile devices like smartphones, iPads, etc. Social media like Facebook, Twitter, WhatsApp, YouTube, LinkedIn help people interact with each other around the world. Social media is also contributing to make this world a better place to live in by raising social issues. Internet of Things (IoT), a network of a large number of objects, computing devices embedded with microchips, sensors, actuators are making this world a smart place to inhabit.

Emerging trends are state-of-the-art technologies which gain popularity and set a new trend among users. In this chapter, we will learn about some emerging trends like cloud computing, mobile computing, social media and Internet of Things (IoT), which are evolving very fast, and also discuss their role and future impact on digital economy and interaction in digital societies.

3.2 ARTIFICIAL INTELLIGENCE (AI)

Artificial Intelligence is a technology that is already impacting how users interact with, and are affected by, the internet and its impact will only continue to grow. AI traditionally refers to an artificial creation of human-like intelligence that can learn, reason, plan, perceive or process natural language. The two words Artificial and Intelligence, where Artificial means 'man-made', and Intelligence means 'thinking power', together mean 'a man-made thinking power'.



Have you ever wondered how maps on your smartphone are able to guide you to take the fastest route to your destination by analyzing real-time data, such as traffic congestion? Or uploading a photo on a social networking site, has it ever happened that your friends in the photograph were recognized and tagged automatically? These are some of the examples of application of Artificial Intelligence. The intelligent digital personal assistants like Siri, Google Assistant, Cortana, Alexa are all powered by AI. Artificial Intelligence endeavours to simulate the natural intelligence of human beings into machines, thus making them behave intelligently. Thus, Artificial Intelligence aims to create Expert Systems, i.e., systems which display intelligent behaviour, learn, demonstrate, explain and advise its users. AI also aims at implementing human intelligence in machines by creating systems that can understand, think, learn and behave like humans.

Applications of AI

The applications of Artificial Intelligence are endless. The technology can be applied to different sectors and industries. AI is being tested and used in the health care industry for dosing drugs, various other treatments, and also for surgical procedures in the operating room.

Apart from the above important applications, AI has been dominant in various other fields such as:

- Gaming
- Natural Language Processing
- Expert Systems
- Vision Systems
- Speech Recognition
- Handwriting Recognition
- Intelligent Robots

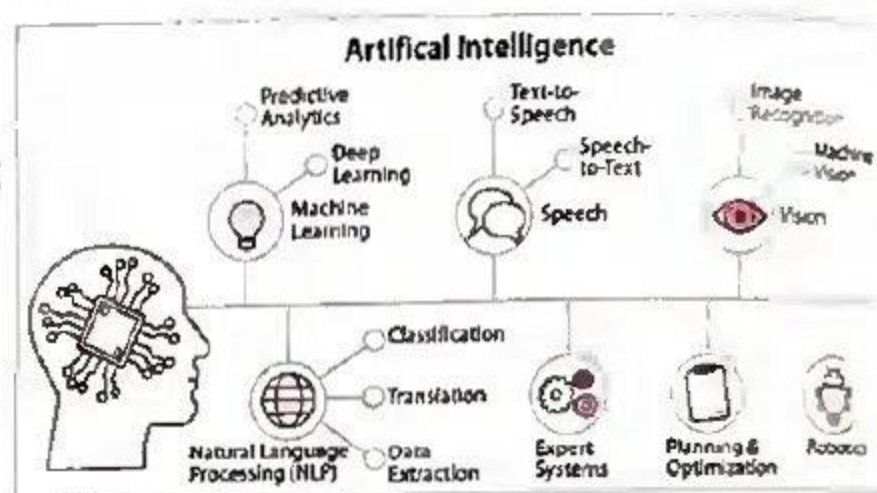


Fig. 3.1: Applications of Artificial Intelligence (AI)

Thus, we may say that Artificial Intelligence is a field of science and technology based on disciplines such as Computer Science, Biology, Psychology, Linguistics, Mathematics and Engineering.

3.2.1 Machine Learning

Machine Learning is an application of Artificial Intelligence (AI). It focuses on providing systems which enable the computer to automatically learn and improve without being explicitly programmed. Learning from the previous tasks handled by the machine leads to better intelligence with better response to new inputs and situations in the future. This is called machine learning.

Machine Learning focuses on development of computer programs that can access data and use it themselves and learn from it.

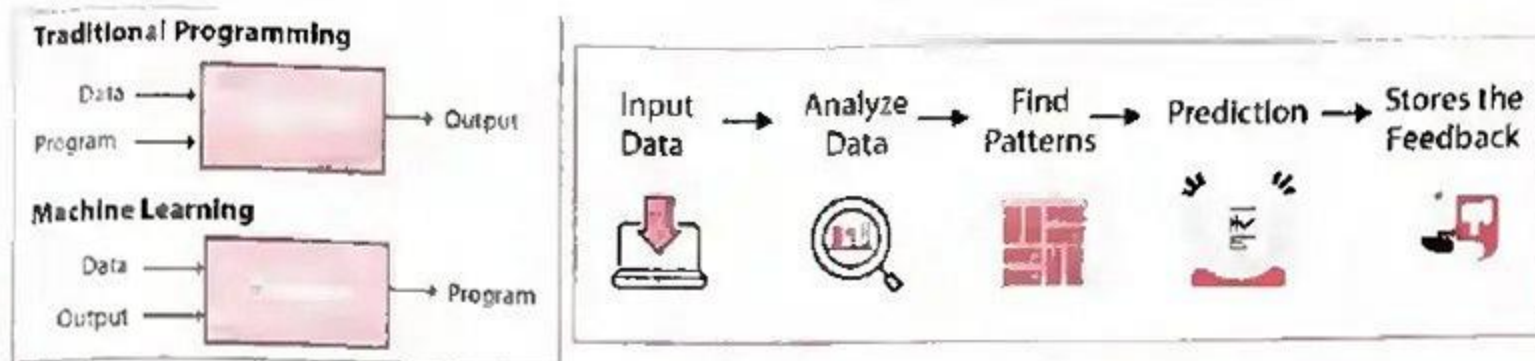


Fig. 3.2: How Machine Learning works

The process of learning begins with observation of data, such as examples, direct experience, or instructions, in order to look for patterns in data and make better decisions in future based on examples that we provide. **The primary aim of Machine Learning is to allow computers learn automatically** without human intervention or assistance and act accordingly.

Machine Learning consists of several algorithms that fetch data to learn on their own and make certain predictions. These algorithms, called models, are first trained and tested using a training and testing data, respectively.

After successive trainings, once these models are able to give results to an acceptable level of accuracy, they are used to make predictions about new and unknown data.

Some of the applications of Machine Learning are:

- Virtual Personal Assistants
- Image Recognition
- Email Spam Filtering
- Text & Speech Recognition
- Medical Diagnosis
- Automatic Translation
- Traffic Prediction
- Online Fraud Detection
- Web Search and Recommendation Engines

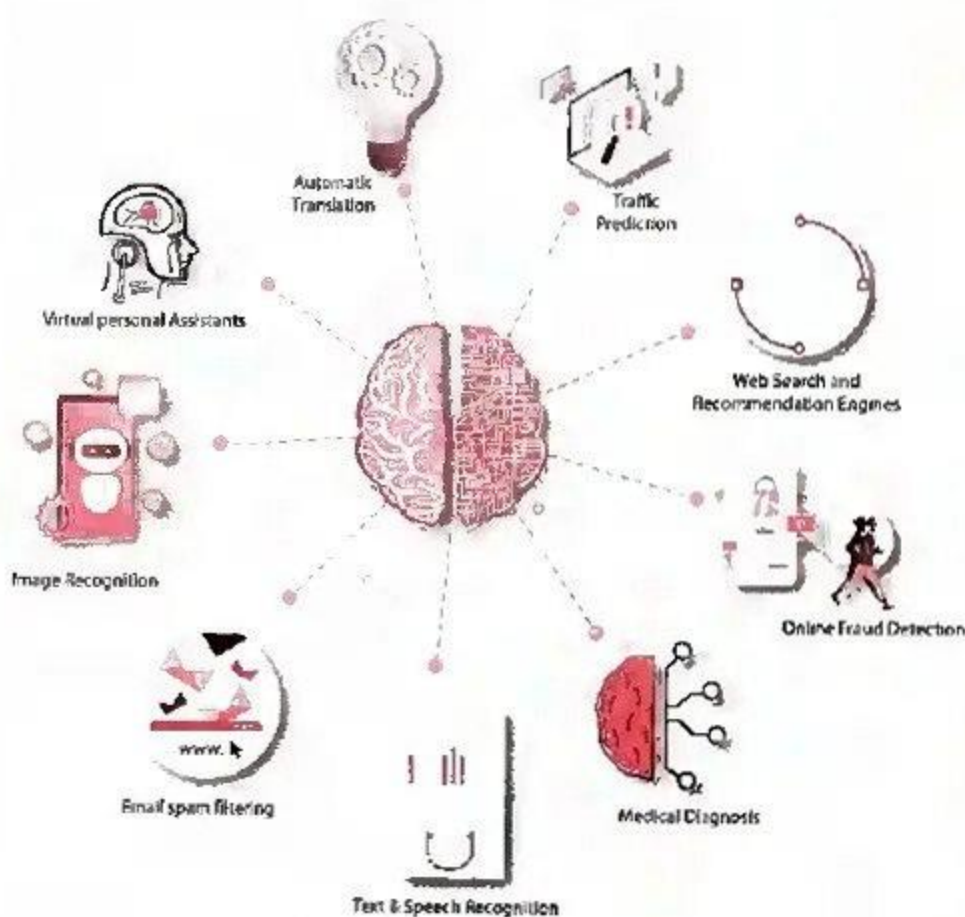


Fig. 3.3: Applications of Machine Learning

3.2.2 Natural Language Processing (NLP)

Many of us already use NLP daily without realizing it. You've probably used at least one of the following tools:

- Spell checker
- Autocomplete
- Spam filters
- Voice text messaging
- Language translation

Natural Language Processing (NLP) is a subfield of Artificial Intelligence that helps computers understand human language.

Nowadays when you say, "Alexa, play this song," a device starts playing that song. This complex interaction could become possible by NLP along with other AI elements such as machine learning and deep learning. NLP makes it possible for computers to read text, hear speech, interpret it, measure sentiment and determine which parts of human language are important. NLP is a branch of Artificial Intelligence that deals with the interaction between computers and humans using the natural language.

Most NLP techniques rely on machine learning to derive meanings from human languages. By NLP and machine learning are subsets of Artificial Intelligence.

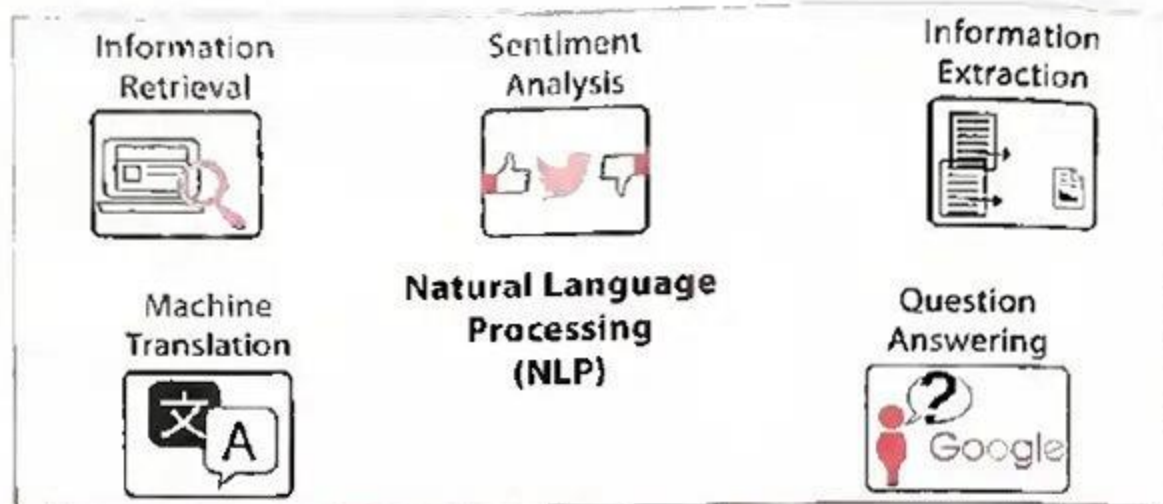


Fig. 3.4: Natural Language Processing (NLP) for Artificial Intelligence

Natural Language Processing is the driving force behind the following common applications

- Language translation applications such as Google Translate.
- Word Processors such as Microsoft Word and Grammarly that employ NLP to check grammatical accuracy of texts.
- Interactive Voice Response (IVR) applications used in call centres to respond to certain users' requests.
- Personal assistant applications such as Google Assistant, Siri and Alexa.

In fact, it is possible to search the web or operate or control our devices using our voice. All this has been made possible by NLP. An NLP system can perform text-to-speech and speech-to-text conversion as shown in Fig. 3.5.



Fig. 3.5: Application of Natural Language

Machine translation is an emerging field where machines are able to translate text from one language to another with a fair amount of correctness. Another emerging application area is automated customer service where computer software can interact with customers to resolve their queries or complaints.

Thus, the field of NLP involves making computers perform useful tasks with the natural languages that humans use. The input and output of an NLP system can be:

- Speech
- Written text

3.2.3 Immersive Experiences

An “immersive experience” pulls a person into a new or augmented reality, enhancing everyday life via technology. It often uses one or more technologies linked together.

With three-dimensional (3D) videography, the joy of watching movies in theatres has reached a new level. Video games are also being developed to provide immersive experience to the player. Immersive experience allows us to visualize, feel and react by stimulating our senses. It enhances our interaction and involvement, making them more realistic and engaging.

Immersive experiences have been used in the field of training, such as driving simulators (Fig. 3.6), flight simulators and so on. Immersive experience can be achieved using virtual reality and augmented reality.

Virtual Reality (VR)

Virtual Reality (VR) is the use of computer technology to create a virtual simulator, away from real environment. Unlike traditional user interfaces, VR puts the user inside an experience. Instead of viewing a screen in front of them, users are immersed and able to interact with 3D worlds. By simulating as many senses as possible, such as vision, hearing, touch, even smell, the computer is transformed into a gatekeeper to this artificial world.

Everything that we experience in our reality is perceived through our senses. From this came the idea that if we can present our senses with made-up or non-real information, our perception of reality would also alter in response to that.

At present, it is achieved with the help of VR headsets. In order to make the experience of VR more realistic, it promotes other sensory information like sound, smell, motion, temperature, etc. It is a comparatively new field and has found its applications in gaming (Fig. 3.7), military training, medical procedures, entertainment, social science and psychology, engineering and other areas where simulation is needed for a better understanding and learning.



Fig. 3.6: Driving Simulator



Fig. 3.7: VR Headset

Augmented Reality (AR)

Augmented Reality (AR) is one of the biggest technology trends today and it is only going to get bigger as AR-ready smartphones and other devices become more accessible around the world. AR lets us see real-life environment right in front of us—trees swaying in the park, dogs chasing balls, kids playing soccer—with a digital augmentation overlaid on it.

This superimposition of computer-generated perceptual information over the existing physical surroundings is called Augmented Reality. It adds components of the digital world to the physical world, along with the associated tactile and other sensory requirements, thereby making the environment interactive which can be digitally manipulated. AR adds digital elements to a live view often by using the camera in a smartphone.

Examples of Augmented Reality experiences include Snapchat lenses and the global gaming app PokémonGO. In other words, if you see the real world supplemented with digital objects, that is AR. Imagine you want to buy a piece of furniture—let us say a chair. Augmented reality technology can help you check how different chairs will look in your room and pick the one that fits best.

IKEA is the first company to use augmented reality which allows you to select anything from the store's catalogue and see how a certain piece of furniture will fit in your space.



Mixed Reality (MR)

A mixed-reality environment goes a step beyond augmented reality because users can interact in real time with virtual objects that are placed within the real world. These virtual items will respond and react to users as if they were actual objects.

In order to experience mixed reality, an MR headset is used. MR headsets offer a holographic experience through translucent glasses or an immersive experience. Some mixed reality products that are currently available include Microsoft's HoloLens, Acer Windows Mixed Reality, Lenovo Explorer, and Samsung Odyssey. It is the gesture/gaze/voice recognition technology through a pair of motion controllers or through the MR headset that helps deliver a believable mixed-reality experience. It takes a lot more processing power to enable a mixed-reality experience than it does for a virtual or augmented reality experience.



Fig. 3.8: MR Headset

3.2.4 Robotics

Robotics is an interdisciplinary research area at the interface of computer science and engineering. Robotics involves design, construction, operation and use of robots. The goal of robotics is to design intelligent machines that can help and assist humans in their day-to-day lives and keep everyone safe.

Robotics helps develop machines that can substitute for humans and replicate human actions. Robots can be used in many situations and for several purposes, but today many are being used in hazardous environments (including inspection of radioactive materials, bomb detection and deactivation), manufacturing processes, or where humans cannot survive (e.g., in space, under water, in high temperature).



Fig. 3.9: Robotics

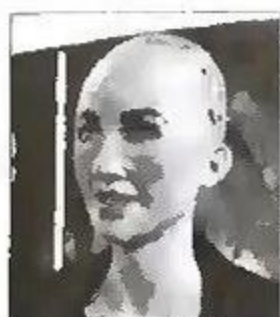
Robots that resemble humans are known as **humanoids**. Robots are being used in industries, medical science, bionics, scientific research, military, etc. Some examples are:

Sophia is a realistic humanoid robot capable of displaying human-like expressions and interacting with people. It is designed for research, education and entertainment. It is the world's first **robot citizen** and the first **robot Innovation Ambassador** for the United Nations Development Programme.

NASA's Mars Exploration Rover (MER) mission is a robotic space mission to study planet Mars.

Mitra is the first Indian-made humanoid that uses artificial intelligence, visual data processing, facial recognition and also imitates human gestures and facial expressions.

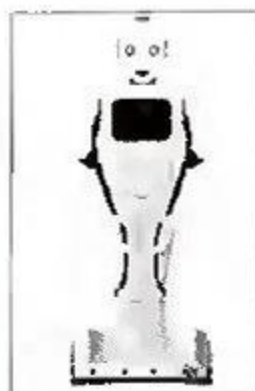
A drone is an unmanned aircraft which can be remotely controlled or can fly autonomously through software-controlled flight plans in their embedded systems, working in conjunction with onboard sensors and GPS.



Sophia



Rover



Mitra



Drone

Drones are being used in many fields, such as journalism, filming and aerial photography, shipping or delivery at short distances, disaster management, search and rescue operations, health care, geographic mapping and structural safety inspections, agriculture, wildlife monitoring or poaching, besides law enforcement and border patrolling.

3.3 BIG DATA

The term Big Data refers to a huge volume of data that cannot be stored and processed by any traditional data storage or processing units. Big Data is generated on a very large scale and used by many multinational companies to process and analyze in order to uncover insights and improve the business of many organizations.

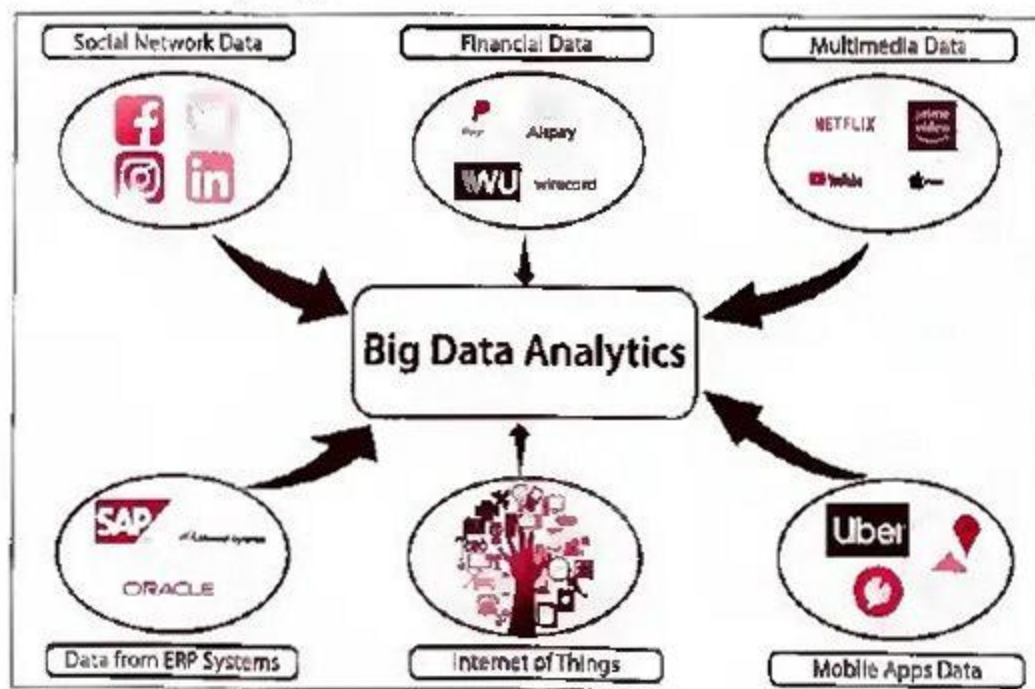


Fig. 3.10: Big Data

Analysis of big data allows analysts, researchers and business users to make better and faster decisions using data that was previously inaccessible or unusable. Businesses can use advanced analytics techniques such as text analytics, machine learning, predictive analytics, data mining, statistics and natural language processing to gain new insights from previously untapped data sources independently or together with existing enterprise data.

Big data cannot be processed and analyzed using traditional data processing tools as the data is not only voluminous but also unstructured, e.g., our posts, instant messages and chats, photographs that we share through various sites, our tweets, blog articles, news items, opinion polls and their comments, audio/video chats, etc.

3.3.1 Characteristics of Big Data

Big data is distinguishable from traditional data on the basis of the following five important characteristics as shown in Fig. 3.11.

(a) Volume

Volume is one of the characteristics of big data. We already know that big data indicates huge 'volumes' of data that is being generated on a daily basis from various sources like social media platforms, business processes, machines, networks, human interactions, etc. Such a large amount of data is stored in data warehouses.

(b) Velocity

Velocity essentially refers to the speed at which data is being created in real time. In a broader prospect, it comprises the rate of change, linking of incoming data sets at varying speeds and activity bursts.

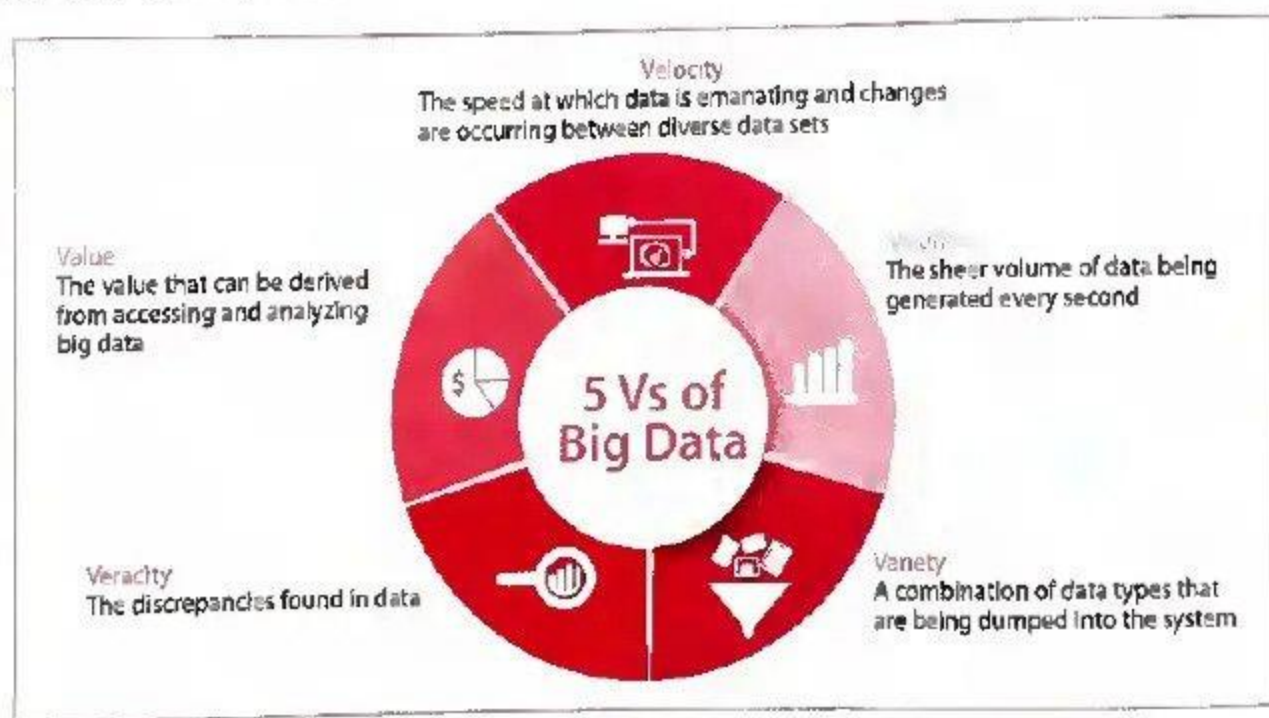


Fig. 3.11: Characteristics of Big Data

(c) Variety

Variety of big data refers to structured, unstructured and semi-structured data that is gathered from multiple sources. While in the past, data could only be collected from spreadsheets and databases, today data comes in an array of forms such as emails, PDFs, photos, videos, audios, SM posts, so on and so forth. Variety is one of the important characteristics of big data.

One of the major implementations of sensors that we observe is when you hold your mobile vertically or horizontally. The display also changes to vertical or horizontal with respect to the way we hold our mobile. This is possible with the help of two sensors, namely accelerometer and gyroscope (gyro). The accelerometer sensor in the mobile phones detects the orientation of the phone. The gyroscope sensor tracks rotation or twist of your hand and adds to the information supplied by the accelerometer.

Sensors are commonly used for monitoring and observing elements in real-world applications. The evolution of smart electronic sensors is contributing in a big way to the evolution of IoT which will lead to creation of new sensor-based, intelligent systems.

3.4.3 Smart Cities

A Smart City is an urban area that uses different types of electronic IoT sensors to collect data and then use insights gained from that data to manage assets, resources and services efficiently. It includes data collected from citizens, devices and assets that is processed and analyzed to monitor and manage traffic and transportation systems, power plants, utilities, water supply networks, waste management, crime detection, information systems, schools, libraries, hospitals and other community services.



Fig. 3.13: Smart City

A smart city makes use of computer and communication technology along with IoT, WoT to manage and distribute resources efficiently. Smart buildings shown in Fig. 3.13 use sensors to detect earthquakes and then warn nearby buildings so that they can prepare themselves accordingly. A smart bridge uses wireless sensors to detect any loose bolt, cable or crack. It alerts concerned authorities through SMS. A smart tunnel also uses wireless sensors to detect any leakage or congestion in the tunnel. This information can then be sent as wireless signals across the network of sensor nodes to a centralized computer system for further analysis.

Every sphere of life in a city like transportation systems, power plants, water supply networks, waste management, law enforcement, information systems, schools, libraries, hospitals and other community services works in unison to optimize the efficiency of city operations and services.

3.5 CLOUD COMPUTING

Cloud computing is a technology of distributed data processing in which some scalable information resources and capacities are provided as a service to multiple external customers through the internet technology. It allows storing, accessing data and programs using the internet.

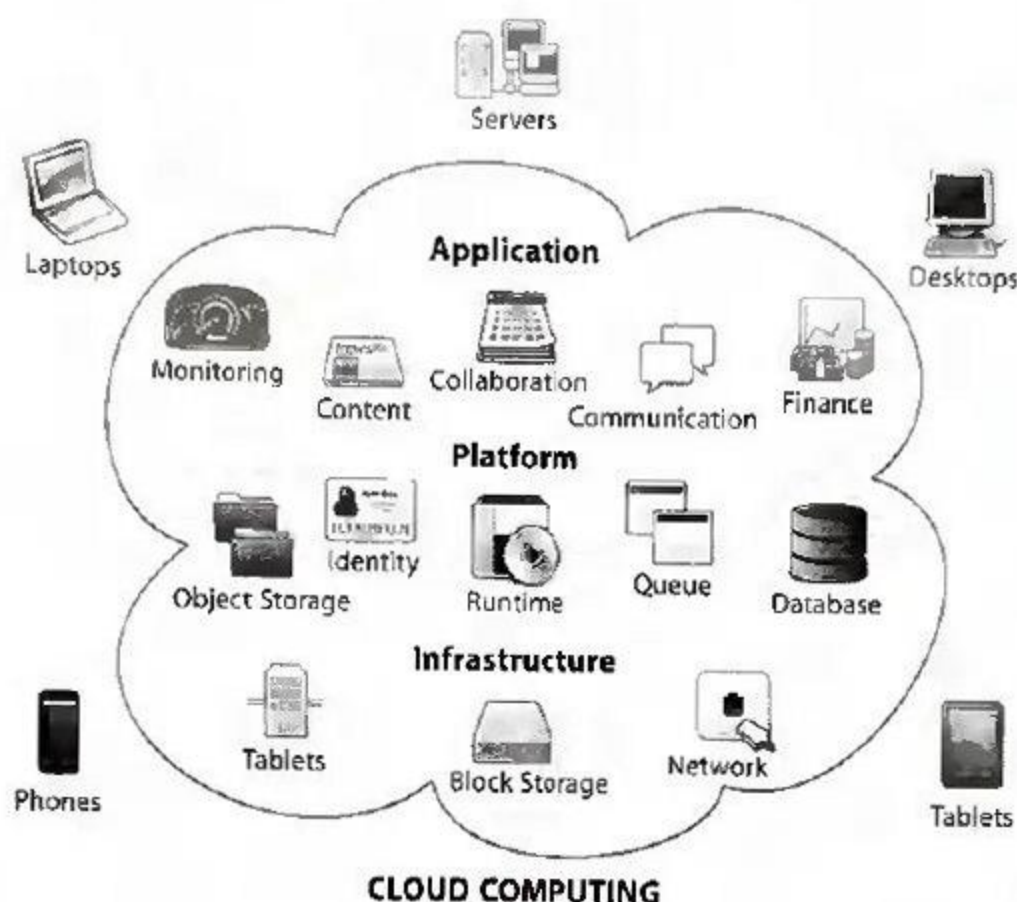


Fig. 3.14: Concept of Cloud Computing

There are broadly three types of clouds one can access:

- (a) Public Cloud
 - (b) Private Cloud
 - (c) Hybrid Cloud
- (a) **Public Cloud:** It is defined as a public or common cloud service provided to multiple users on a network. Public cloud services are provided through a common portal or virtual platform owned and operated by a third-party cloud provider. In a public cloud service, resources such as memory, hardware devices and network devices are shared by all the clients (or organizations that use the same cloud called subscribers), known as cloud **tenants**, such as Google Drive, Dropbox, Microsoft OneDrive, iCloud, Amazon Cloud Drive.
- (b) **Private Cloud:** Private cloud is a virtual private interface provided by an individual or owned by one organization. In a private cloud, all the resources such as memory and services are dedicated solely to an organization. Some companies also pay third-party service providers to host their private cloud.

- (c) **Hybrid Cloud:** Hybrid cloud combines public and private clouds together by technology that allows data and applications to be shared between them. It provides mixed computing, storage and services environment made up of on-premises infrastructure such as Amazon Web Services (AWS) and Microsoft Azure.

Cloud computing further consists of services such as:

- **IaaS (Infrastructure as a Service)**—A computer infrastructure, typically presented in the form of physical and virtualized resources that provide consumers with the basic building blocks needed to run applications and workloads in the cloud. The IaaS providers can offer different kinds of computing infrastructure, such as servers, virtual machines (VM), storage and back-up facility, network components, operating systems or any other hardware or software. It is a service within the concept of cloud hosting. Examples of IaaS are: Windows Azure, Rackspace, Google Computing Engine, etc.
- **PaaS (Platform as a Service)**—An integrated platform for the development, deployment, testing and support of web or mobile applications. PaaS is designed to make it easier for the developers to quickly create apps without worrying about managing the underlying infrastructure of servers, storage, network and databases needed for development. It is presented as a service on the basis of the concept of cloud hosting. Examples of PaaS are: AWS Elastic Beanstalk, Heroku, Windows Azure, etc.
- **SaaS (Software as a Service)**—The business model of software licence and software upgrades which involves development and support of the software vendor. Customers also have the opportunity of paid use of it, through the internet, usually with a web browser on their phone, tablet or PC. Examples of SaaS are: Microsoft Office365, Google Workspace, Dropbox, Cisco WebEx, GoToMeeting, etc.

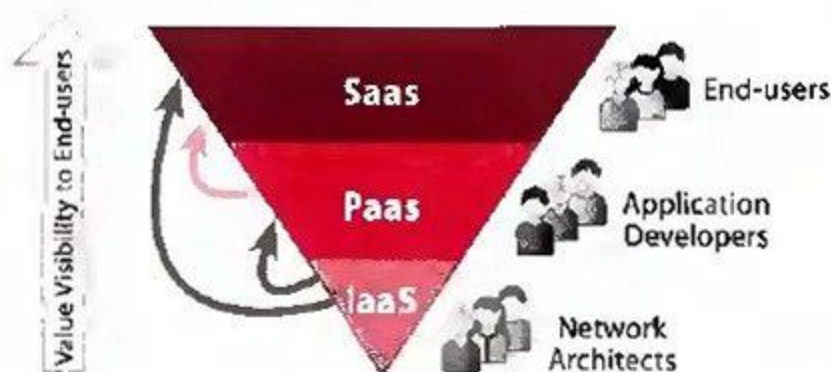


Fig. 3.15: Implementation of Cloud Computing

- **DaaS (Desktop as a Service)**—Another business model licence, this software, first introduced in the early 2000s, is a slightly improved model of SaaS. It mostly involves use of multiple services at the same time necessary to complete a work.

In addition to the above, within the concept of cloud hosting technology, there are also common notion Data as a Service and Everything as a Service, respectively. Both concepts show that through the World Wide Web using Cloud Hosting, any requirement can be met in the processing of information. This is the main advantage of cloud computing hosting in the IT-based business solutions.

CTM: Cloud computing refers to having access to all your applications and data from any network device.

3.6 GRID COMPUTING

Grid Computing can be defined as a network of computers working together to perform a task that would be rather difficult for a single machine. All machines on that network work under the same protocol to act like a virtual supercomputer. The task that they work on may include analyzing huge data sets or simulating situations which require high computing power. Computers on the network contribute resources like processing power and storage capacity to the network.

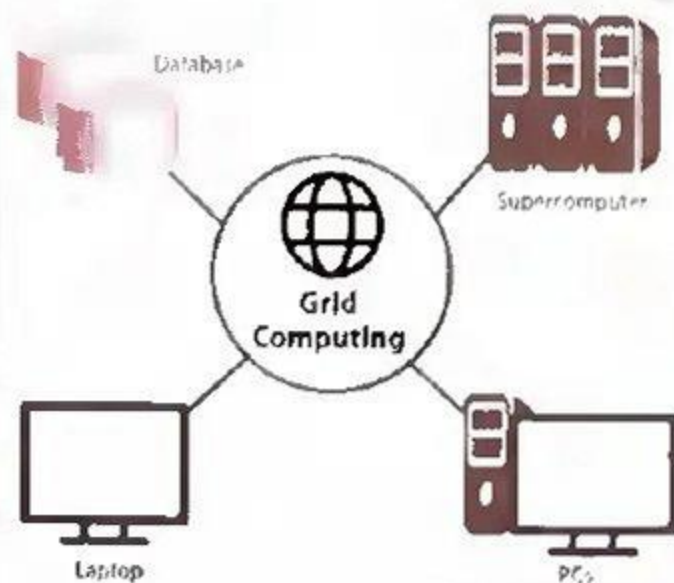


Fig. 3.16: Grid Computing Framework

Grid Computing is a subset of distributed computing, where a virtual supercomputer consists of machines on a network connected by some bus, mostly Ethernet, or sometimes the Internet. It can also be seen as a form of Parallel Computing where, instead of many CPU cores on a single machine, it contains multiple cores spread across various locations.

Working

A Grid Computing network mainly consists of the following three types of machines:

1. **Control Node:** A computer, usually a server or a group of servers, which administers the whole network and keeps the account of the resources in the network pool.
2. **Provider:** The computer which contributes its resources to the network resource pool.
3. **User:** The computer that uses the resources on the network.

Grid can be of two types—(i) Data grid, used to manage large and distributed data having the required multi-user access, and (ii) CPU or Processor grid, where processing is moved from one PC to another as needed or a large task is divided into sub-tasks and allotted to various nodes for parallel processing.

Grid Computing is different from IaaS cloud service. In case of IaaS, there is a service provider who rents the required infrastructure to the users, whereas in grid computing, multiple computing nodes join together to solve a common computational problem. To set up a grid by connecting numerous nodes in terms of data as well as CPU, a middleware is required to implement the distributed processor architecture. The Globus toolkit (<http://toolkit.globus.org/toolkit>) is one such software toolkit used for building grids and it is open source. It includes software for security, resource management, data management, communication, fault detection, etc.

3.7 BLOCKCHAIN TECHNOLOGY

Blockchain technology works on the concept of decentralized and shared database where each computer has a copy of the database. A block can be considered as a secured chunk of data or valid transaction. Each block has some data called its header, which is visible to every other node, while only the owner has access to the private data of the block. Such blocks form a chain called blockchain (Fig. 3.17).

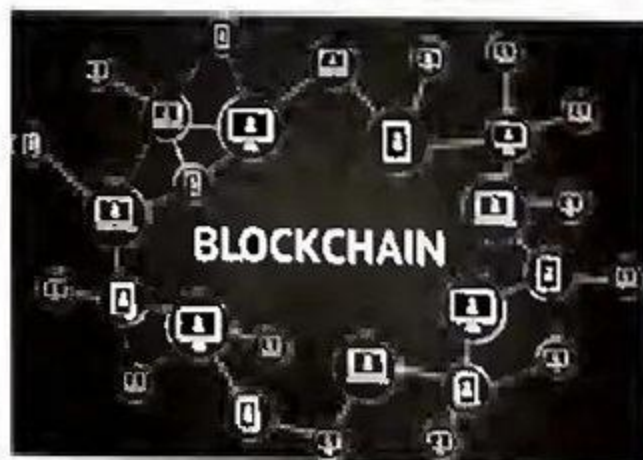
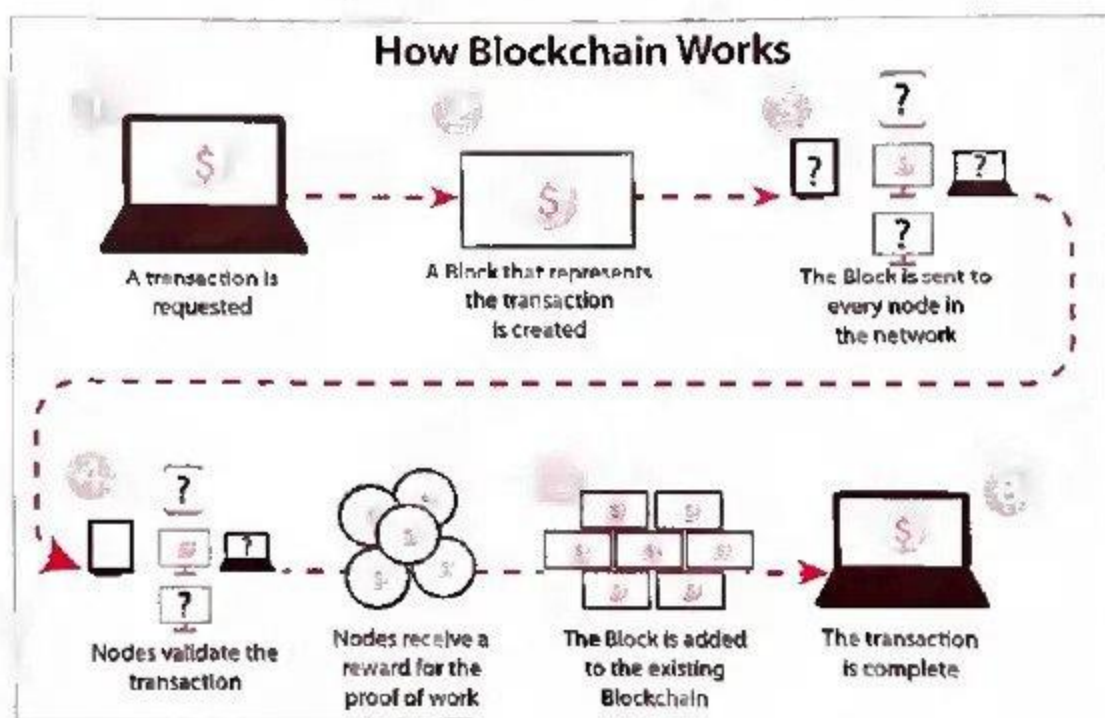


Fig. 3.17: Blockchain Technology

A simple analogy for understanding blockchain technology is a Google Doc. When we create a document and share it with a group of people, the document is distributed instead of being copied or transferred. This creates a decentralized distribution chain that gives everyone access to the document at the same time. No one is locked out awaiting changes from another party, while all modifications to the document are being recorded in real time, making changes completely transparent.

We can define blockchain as a system that allows a group of connected computers to maintain a single updated and secure ledger. Each computer or node that participates in the blockchain receives a full copy of the database. It maintains an 'append only' open ledger which is updated only after all the nodes within the network authenticate the transaction. Safety and security of the transactions is ensured because all members in the network keep a copy of the blockchain and so it is not possible for a single member of the network to make changes or alter data.

The most popular application of blockchain technology is in digital currency. However, due to its decentralized nature with openness and security, blockchains are being seen as one of the ways to ensure transparency, accountability and efficiency in business as well as in governance systems. For example, in health care, better data sharing between health care providers would result in high probability of accurate diagnosis, more effective treatment and the overall increased ability of health care organizations to deliver cost-effective care.



Apart from the above implementation areas, the blockchain technology can be used in diverse sectors such as voting system, banking, media, telecom, travel and hospitality, etc.



MEMORY BYTES

- Emerging trends are the state-of-the-art technologies which gain popularity and set a new trend among users.
- Artificial Intelligence endeavours to simulate the natural intelligence of human beings into machines, thus making them intelligent.
- Machine learning comprises algorithms that use data to learn on their own and make predictions.
- Natural Language Processing (NLP) facilitates communication with intelligent systems using a natural language.
- Virtual reality allows a user to look at, explore and interact with virtual surroundings, just like one can do in the real world.

- Cloud computing, often referred to as “the cloud”, is the delivery of on-demand computing resources over the internet on a pay-for-use basis.
- The superimposition of computer-generated perceptual information over the existing physical surroundings is called augmented reality.
- Robotics can be defined as a science, primarily associated with the design, fabrication, theory and application of robots.
- Big data holds rich information and knowledge which can be of high business value. Five characteristics of Big data are: Volume, Velocity, Variety, Veracity and Value.
- Data analytics is the process of examining data sets in order to draw conclusions about the information they contain.
- Internet of Things (IoT) is a network of devices that have an embedded hardware and software to communicate (connect and exchange data) with other devices on the same network.
- A sensor is a device that takes input from the physical environment and uses built-in computing resources to perform pre-defined functions upon detection of specific input and then processes data before passing it on.
- Blockchain technology uses a shared database of chained blocks where copies of database exist on multiple computers.

OBJECTIVE TYPE QUESTIONS

1. Fill in the blanks.

- (a) refers to the ability of machines to perform cognitive tasks like thinking, perceiving, learning, problem-solving and decision-making.
- (b) NLP is a branch of AI that facilitates between humans and computers.
- (c) Using a, users experience being transported into real and imagined worlds.
- (d) The goal of is to design intelligent machines that can help and assist humans in their day-to-day lives and keep everyone safe.
- (e) In, data is not stored on the PC but it is available and accessible across devices.
- (f) is a cloud computing service that allows users to create and deploy applications.
- (g) A sensor converts the into a signal which can be measured electrically.
- (h) The “block” in blockchain technology refers to stored in a data structure referred to as “chain”, which is called
- (i) provides on-demand access to application software, usually requiring a licence or subscription by the user.
- (j) The works on the concept of decentralised and shared database where each computer has a copy of the database.
- (k) technology is able to take an existing environment and add a layer of virtual information on top of it.

2. State whether the following statements are True or False.

- (a) AI systems cannot help in decision-making.
- (b) NLP is a trait of Deep Learning, which is an implementation of AI.
- (c) Virtual Reality (VR) is a three-dimensional, computer-generated situation that simulates the real world.
- (d) Cloud-based storage makes use of local computer storage for saving files and information.
- (e) Drone is an example of Augmented Reality.
- (f) CCTV cameras installed at offices, houses, business establishments are a result of Internet of Things (IoT) implementation.
- (g) Smart cities are not sustainable and liveable.
- (h) A grid is a computer network of geographically dispersed and heterogeneous computational resources.

3. Multiple Choice Questions (MCQs)

- (a) IoT stands for
- (i) Industry of Things
 - (ii) Internet of Things
 - (iii) Intelligence of Things
 - (iv) Internal Internet of Things
- (b) Which one of these is **not** an area of AI?
- (i) Computer vision/Image recognition
 - (ii) Voice recognition
 - (iii) Web design
 - (iv) Robotics
- (c) "Virtual reality" is described as:
- (i) Where parts of a person's body are used for identification purposes. They are unique.
 - (ii) A software environment that recreates a real world or imaginary scenario like a game.
 - (iii) The use of computers to encrypt data so that messages can be sent with complete privacy.
 - (iv) Where images are made to appear three-dimensional and have 'depth'.
- (d) Identify the correct print() statement:
- (i) Training soldiers in combat using battlefield simulations
 - (ii) In self-driving cars that do not require a driver
 - (iii) To allow access to buildings
 - (iv) To create prototypes of cars or any object
- (e) Where is virtual reality used?
- (i) Training soldiers in combat using battlefield simulations
 - (ii) In three-dimensional (3D) films
 - (iii) To allow access to buildings
 - (iv) To create prototypes of cars or any object
- (f) Where is computer-assisted translation used in everyday life?
- (i) Training TV hosts
 - (ii) Spell-checkers used in word processing software
 - (iii) Dictation tests
 - (iv) Recording weather conditions
- (g) Which of the following is a positive impact of artificial intelligence in driverless cars?
- (i) An immersive video experience for the viewer using special glasses.
 - (ii) Data is completely secure and cannot be tampered with.
 - (iii) This method ensures that only authorized people can enter restricted buildings or rooms.
 - (iv) Safer transport as there would be fewer road accidents.
- (h) What hardware is typically used for virtual reality?
- (i) A powerful computer, headset with screen to see, headphones to hear and sometimes gloves.
 - (ii) Keyboard, mouse, printer and plotter.
 - (iii) Smartphone, mic and tablet.
 - (iv) Plotters, mouse, speakers and tablet.
- (i) A company interested in cloud computing is looking for a provider who offers a set of basic services such as virtual server provisioning and on-demand storage that can be combined into a platform for deploying and running customized applications. What type of cloud computing model fit these requirements?
- (i) Platform as a Service
 - (ii) Software as a Service
 - (iii) Infrastructure as a Service
 - (iv) None of these
- (j) Which of the following is not a feature of IoT devices?
- (i) Remotely controllable
 - (ii) Security
 - (iii) Can turn themselves off, if necessary
 - (iv) Wearables

SOLVED QUESTIONS

1. Define knowledge base.

Ans. A knowledge base is a store of information consisting of facts, assumptions and rules which an AI system can use for decision-making.

2. How does NLP help differently-abled persons?

Ans. NLP is the machine's ability to process what was said, structure the information received, determine the necessary response and respond in a language that we understand. It also has a key feature of word embedding to help the differently-abled to work with ease.

3. Describe the benefits of cloud computing.

Ans. The benefits of cloud computing are:

- (a) Easy to maintain.
- (b) Increased security at a much lesser cost.
- (c) On-demand self-service by consumers.
- (d) Rapid scaling of capacity.
- (e) Resource pooling of physical and virtual resources.

4. Define the term Robotics.

Ans. Robotics is an interdisciplinary branch of technology requiring applications of mechanical engineering, electronics and computer science, among others. Robotics is primarily concerned with the design, fabrication, operation and application of robots.

5. Briefly explain the role played by robots in the medical field.

Ans. Prosthetic robotic limbs interface with the nervous system to restore movement and sense of touch to amputees. Automated dispensing robots limit medication-dispensing errors in pharmacy settings. Clinical training robots offer lifelike simulations of medical conditions to help doctors during their training.

6. Drone—a utility of Robotics—can be helpful in the event of a natural calamity. How?

Ans. Yes, drones can be of great help during natural calamities. Following a disaster, drones can help relief workers understand the big picture, locate survivors more quickly, perform structural analysis of damaged infrastructure, deliver supplies and equipment, and evacuate casualties and help put out fires.

7. Define the term big data. Which all activities can contribute to big data?

Ans. Big data is a term used to describe a collection of data that is huge in volume and still growing exponentially. In short, such data is so large and complex that none of the traditional data management tools are able to store it or process it efficiently. The activities such as social messaging and posts, instant messages and chats, photographs that we share through various sites, our tweets, blogs, news items, opinion polls and their comments, audio/video chats, etc. contribute to big data.

8. Explain the types of Grids.

Ans. Grid can be of two types—(i) Data grid, used to manage large and distributed data having the required multi-user access, and (ii) CPU or Processor grid, where processing is moved from one PC to another as needed or a large task is divided into subtasks, and allotted to various nodes for parallel processing.

9. Name some areas where the concept of block chain technology can be useful.

Ans. Few areas of block chain technology are:

- (i) Payment processing and money transfers
- (ii) Monitor supply chains
- (iii) Digital IDs
- (iv) Data sharing
- (v) Copyright and royalty protection
- (vi) Digital voting
- (vii) Medical recordkeeping
- (viii) Managing Internet of Things (IoT) networks

10. Explain the areas where Natural Language Processing (NLP) is used extensively.

Ans. Natural Language Processing is the driving force behind the following common applications.

- Language translation applications such as Google Translate
- Word Processors such as Microsoft Word employ NLP to check grammatical accuracy of texts
- Interactive Voice Response (IVR) applications used in call centres to respond to certain user requests
- Personal assistant applications such as Google Assistant, Siri and Alexa.

11. Differentiate between Cloud Computing and Grid Computing.

Ans.

	Cloud Computing	Grid Computing
1.	Cloud computing is a client-server computing architecture where resources located at remote locations are made available to anyone from anywhere.	Grid computing is a distributed architecture where a large number of computers are connected parallelly to solve a given problem.
2.	In cloud computing, resources are centrally managed.	In grid computing, resources are managed on collaboration pattern.
3.	In cloud computing, the users pay for the use.	In grid computing, the users do not pay for use.
4.	In cloud computing, cloud servers are owned by infrastructure providers.	In grid computing, grids are owned and managed by the organization.
5.	Cloud computing is a highly accessible service.	Grid computing is low on accessibility as compared to cloud computing.

12. What is Machine Learning? Write its applications.

Ans. Machine Learning is an application of Artificial Intelligence (AI). It focuses on providing systems which enable the computer to automatically learn and improve without being explicitly programmed.

Machine Learning focuses on development of computer programs that can access data and use it themselves and learn from it. Some of the applications of machine learning are:

- (i) Image recognition
- (ii) Text & speech recognition
- (iii) Online fraud detection
- (iv) E-mail spam filtering
- (v) Virtual personal assistant

13. How is Virtual Reality different from Augmented Reality?

Ans. **Virtual Reality (VR):** A fully immersive experience where a user leaves the real-world environment behind to enter a fully digital environment via VR headsets.

Augmented Reality (AR): An experience where virtual objects are superimposed onto the real-world environment via smartphones, tablets, heads-up displays or AR glasses.

14. Differentiate between Public Cloud and Private Cloud.

Ans. Public Cloud services are owned and operated by a third party cloud service providers which deliver their computing resources like servers and storage over the Internet.

Private Cloud refers to cloud computing resources used exclusively by a single business or organization.

15. What do you understand by Internet of Things? List some of its potential applications.

Ans. IoT or Internet of Things refers to connecting many devices through hardware or software and communicate with each other as well as human beings through internet to work in collaboration and assist each other and form a networking of things.

Potential Applications of IoT:

- (i) Smart Home
- (ii) Wearables
- (iii) Smart City
- (iv) Smart Grids
- (v) Industrial Internet

16. Justify the statement: 'Storage of data is cost-effective and time-saving in cloud computing.' [HOTS]

Ans. Cloud computing stores data into a server and it is available almost free of cost or at a nominal cost. When it comes to time-saving by saving data on the internet, there is no need to wait to start up any computer or device. It is easily available with a browser program or an app. So, it can be accessed any time from anywhere.

17. Write examples of the following:

- (i) Government-provided cloud computing platform
- (ii) Large-scale private cloud service providers and the services they provide

Ans. (i) The Government of India has provided a government cloud service "GI Cloud" which has been named 'MeghRaj'.

- (ii) Google is a large-scale private cloud service provider which provides storage, web services and many other things.

18. If the government plans to make a smart school by applying IoT concepts, how can each of the following be implemented in order to transform a school into IoT-enabled smart school?

Ans. IoT-enabled smart school can be implemented by incorporating the following services:

- (i) **e-Textbooks:** e-Textbooks can be made available in e-library by using web services.
- (ii) **Smart Boards:** Smart boards can be operated by digital stylus or pen and should be connected with network.
- (iii) **Online Tests:** A web portal or online examination should be prepared to conduct exams smoothly.
- (iv) **Wi-Fi sensors and classrooms doors:** Wi-Fi sensors and classroom doors can be used to lock and unlock them. These sensors also provide security to students and other stakeholders.
- (v) **Sensors in buses to monitor their location:** These sensors enable quick reporting and monitoring to the coordinates and parents to know the status of their children and the bus after they leave the school.
- (vi) **Wearables (watches or smart belts) for attendance monitoring:** These devices can be used or implemented for students in such a way that identifies a student and marks the attendance accordingly.

19. How are IoT and WoT related?

Ans. IoT and WoT both can be used on the network to communicate with multiple devices. They can provide a communication medium between various devices, which can be an app, a web service or any network-enabled device.

UNSOLVED QUESTIONS

1. Explain the significance of Artificial Intelligence in today's world.
2. What do you understand by Internet of Things? List some of its potential applications.
3. Write a short note on the following:
 - (a) Cloud Computing
 - (b) Big data and its characteristics
4. Explain the following along with their applications:
 - (a) Artificial Intelligence
 - (b) Machine Learning
5. Differentiate between Cloud Computing and Grid Computing with suitable examples.
6. Bharat Infrastructures (P) Ltd, interested in cloud computing, is looking for a provider who offers a set of basic services such as virtual server provisioning and on-demand storage that can be combined into a platform for deploying and running customized applications. What type of cloud computing model fits the following requirements?
 - Platform as a Service
 - Software as a Service
 - Infrastructure as a Service

7. Home automation is a salient feature of IoT. Explain.
8. What is on-demand service? How is it provided in cloud computing?
9. If a Regional Engineering College plans to make a smart college by applying IoT concepts, how can each of the following be implemented in order to transform a college into an IoT-enabled smart college?
 - e-textbooks
 - Smart boards
 - Online tests
 - Wi-fi sensors on classroom doors
 - Sensors in buses to monitor their location
 - Wearables (watches or smart belts) for attendance monitoring
10. (a) Write any two examples of Augmented Reality.
 (b) Write the name and purpose of any one Indian humanoid.
 (c) Expand the following terms:
 - (i) IoT
 - (ii) WoT

CASE-BASED/SOURCE-BASED INTEGRATED QUESTIONS

1. Five friends plan a start-up. However, they have a limited budget and limited computer infrastructure. How can they avail the benefits of cloud services to launch their start-up?
Ans. They can go for registration with any suitable service provider, either government or private. They can register for infrastructure as a service to avail the facility of cloud storage and deploy applications initially.
2. Governments provide various scholarships to students of different classes. Prepare a report on how block chain technology can be used to promote accountability, transparency and efficiency in the distribution of scholarships?
Ans. As we know that in blockchain technology, a separate ledger is created for individuals and stored in a group of connected computers so that transparency and efficiency automatically improve. The details of the students, who can avail of the scholarships, can be stored in their ledger only as it will not be possible for a single member of the network to make changes or alter data. The transactions will also be more authentic.
3. Suggest the technology which will be implemented to perform the following day-to-day tasks:
 - (i) You got a reminder to take medication
 - (ii) You got an SMS alert that you forgot to lock the door
 - (iii) You got an SMS alert that parking space is available near your block
 - (iv) You turned off your LED TV from your wristwatch**Ans.** (i) Smart Health
 (ii) Home Automation
 (iii) Smart Parking
 (iv) Smart Wearable